



## **Tutorial 03**

### **EE-809 Analog IC Design**

**AC characteristics ( $G_m$ ,  $F_T$  &  $F_{Max}$ ) of MOS Transistors**

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## Contents

|                                                                               |    |
|-------------------------------------------------------------------------------|----|
| Introduction .....                                                            | 4  |
| Theoretical Background.....                                                   | 4  |
| Cadence Environment Setup .....                                               | 5  |
| Creating new Circuit Block.....                                               | 5  |
| A) Schematic Capture for transconductance ( $G_m$ ) .....                     | 5  |
| B) Schematic Capture and Simulation for Transition Frequency ( $F_T$ ): ..... | 9  |
| C) Schematic Capture and Simulation for $F_{max}$ :.....                      | 13 |
| Summary .....                                                                 | 17 |
| Exercise Activities .....                                                     | 18 |
| Activity (3.1) .....                                                          | 18 |
| Activity (3.2) .....                                                          | 18 |
| Activity (3.3) .....                                                          | 18 |
| Activity (3.4) .....                                                          | 18 |



## Introduction

Building upon the DC/ **IV** characterization performed in Tutorial 02, this tutorial shifts focus to the small-signal AC and high-frequency performance of MOSFETs. MOSFET AC characteristics are fundamental to evaluating transistor speed, efficiency, and signal amplification in analog and RF integrated circuits. In this tutorial, you will simulate and analyze key dynamic parameters including transconductance (**G<sub>m</sub>**), unity-gain frequency (**F<sub>T</sub>**), and maximum oscillation frequency (**F<sub>Max</sub>**) using Cadence Virtuoso. Additionally, the impact of transistor biasing, channel length (**L**), and overdrive voltage (**V<sub>ov</sub>**) on these metrics will be explored.

### Objectives

The following in the Cadence Virtuoso design suite will be explored:

- Schematic drawing and editing tool
- ADE (analog design environment) simulation tool
  - o DC analysis and operating point parameter extraction (gm, gm/I<sub>D</sub>)
  - o AC analysis for small signal current gain and FT
  - o Parameter extraction (Gm, Cgd, Rg) to calculate the theoretical FMax
  - o Parametric analysis (Sweeping channel length and bias voltages)
  - o Calculator tool

## Theoretical Background

**1) Transconductance** represents the sensitivity of drain current to small variations in gate voltage:

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS, const}}$$

The transconductance efficiency ratio ( $g_m/I_D$ ) measures how effectively a transistor converts DC current into AC gain. It reaches its peak in weak inversion (subthreshold) and decreases as the device moves into strong inversion.

**2) The Transition fFrequency** (or unity-gain frequency) is the frequency at which the short-circuit small-signal current gain ( $h_{21} = \frac{i_{out}}{i_{in}}$ ) drops to unity (0 dB). It represents the fundamental speed limit of the transistor:

$$f_T \approx \frac{g_m}{2\pi(C_{gg})} \rightarrow \text{Eq (1)}$$

Where **gm** = Transconductance

**C<sub>gg</sub>** = total small-signal capacitance at the gate node, equal to the sum of the gate-to-source ( $C_{gs}$ ), gate-to-drain ( $C_{gd}$ ), and gate-to-bulk ( $C_{gb}$ ) capacitances.

**3) Maximum Frequency (F<sub>max</sub>):** While  $f_T$  measures current gain,  $f_{max}$  is the frequency at which the maximum available power gain drops to unity (0 dB). It incorporates the parasitic gate resistance ( $R_g$ ):

$$f_{max} \approx \sqrt{\frac{f_T}{8\pi R_g C_{gd}}} \rightarrow \text{Eq (2)}$$

Unlike  $f_T$ ,  $f_{max}$  accounts for layout-dependent parasitics like gate resistance, emphasizing the need for multi-finger transistor structures in high-frequency design.



## Cadence Environment Setup

After running Cadence Virtuoso, Library Manager/workspace window will popup. All the files appear on the first column, second column shows the circuit cells in the file and in the last column various attributes of the circuits appear like schematic, layout, symbol, maestro etc. The tutorial will begin with the first step of the design cycle of an electrical circuit; the schematic capture.

## Creating new Circuit Block

1. Create new library (**File** → **New** → **Library**) and type a meaningful name for your library (like **Tutorial\_03\_yourname**).
2. A dialog box will appear, select **Attach existing Libraries** then attach PDK **tsmcN65**, into reference technology and click **Ok**. In the library column select the file just created and go to file and create new Cell view (**File** → **New** → **Cell View**), a dialog box will appear give a meaningful name (like **AC\_CHAR\_NMOS**) then click **OK**, new blank schematic editing window will appear. It is recommended that you work on each task within a separate cellview.

## A) Schematic Capture for transconductance ( $G_m$ )

Tutorial 02 evaluated transconductance ( $g_m$ ) as an isolated parameter, this **guided task** uses the  $g_m/I_D$  **ratio** to measure transconductance efficiency, quantifying how effectively DC bias current ( $I_D$ ) generates  $g_m$ , which dictates both small signal voltage gain ( $A_v \propto g_m$ ) and intrinsic bandwidth ( $f_T \propto g_m/C_{gg}$ ).

The analysis involves sweeping  $V_{GS}$  across three inversion regimes:

**Weak Inversion** ( $V_{GS} < V_{th}$ ): Diffusion-dominated; maximum efficiency, but low speed. sits right in the typical  $15\text{--}20\text{ V}^{-1}$  transition window

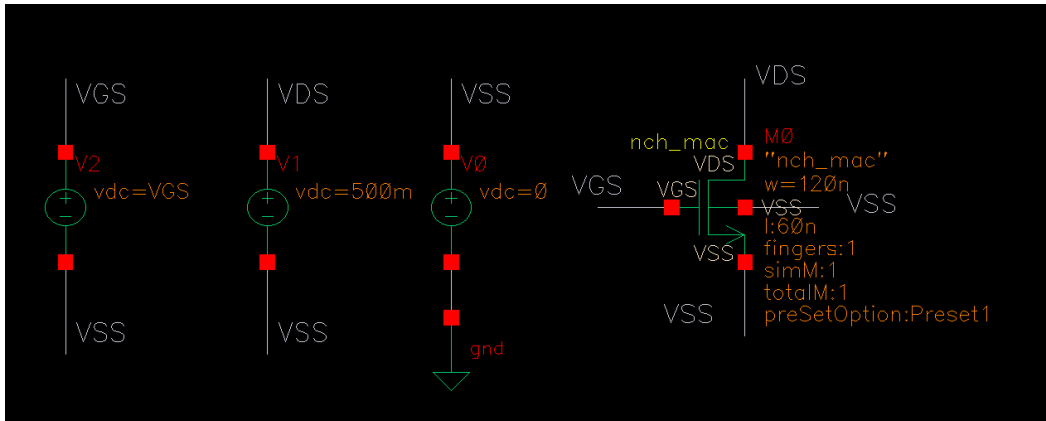
**Moderate Inversion** ( $V_{GS} \approx V_{th}$ ): Transition region; optimal balance of speed and efficiency. Typical  $17\text{--}10\text{ V}^{-1}$  transition window

**Strong Inversion** ( $V_{GS} > V_{th}$ ): Drift-dominated; peak bandwidth at the cost of high power dissipation. Typical  $10\text{--}2\text{ V}^{-1}$  transition window

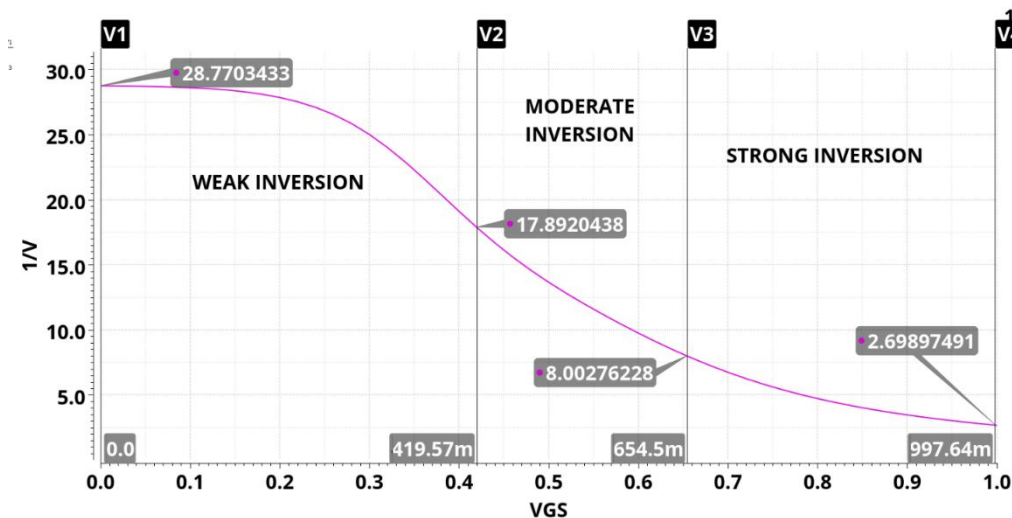
Plotting  $f_T$  against  $g_m/I_D$  illustrates the fundamental trade-off of analog IC design: strong inversion maximizes bandwidth, whereas **moderate inversion** preserves high-frequency speed while significantly reducing power consumption.



1. Make the schematic as shown below



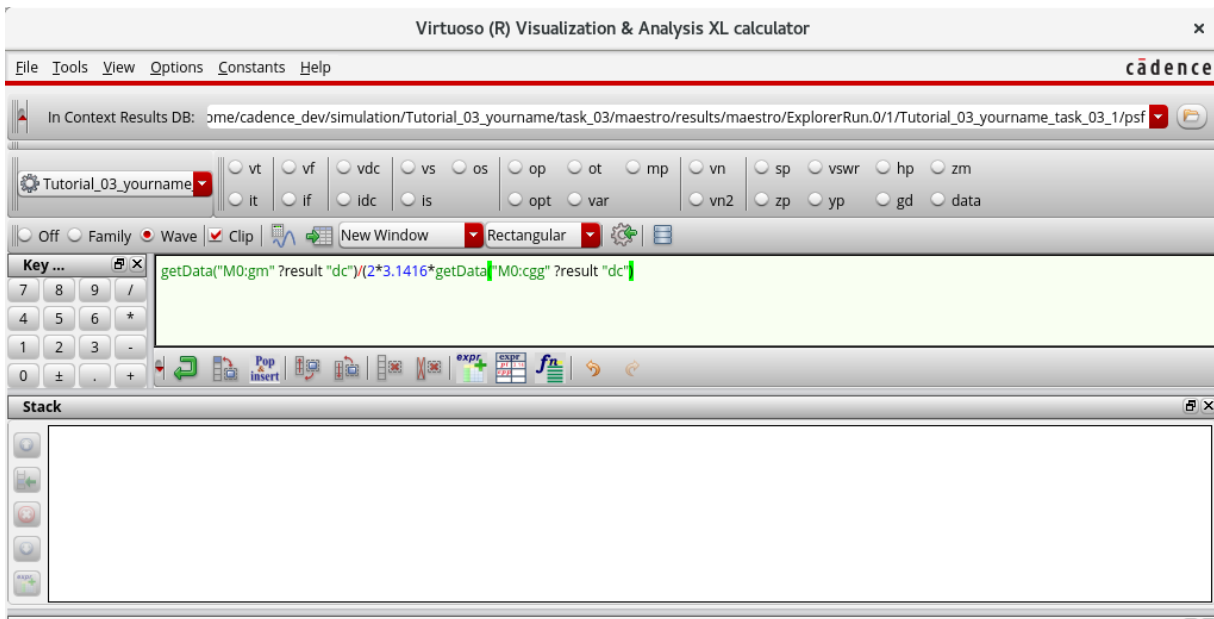
2. **Check and save** the schematic and go to **ADE Explorer**.
3. First copy all the design variables and specify a value of 1 to VGS
4. Specify analyses as DC and select VGS as the design variable and specify its range from 0 to 1.
5. Select the drain current to be plotted.
6. Before simulating, go to **Setup->Simulation files** and under the definition files import the text file "**save\_operating\_points.scs**" created for the previous tutorial. If you don't have this text file, refer to tutorial 02.
7. Next simulate and then go to tools->results browser. Under dc, look for gm, cgg and gm/id. Plot all of these and send their expressions to ADE.
8. The gm/id versus VGS curve will look as shown below. Label your graph as shown.



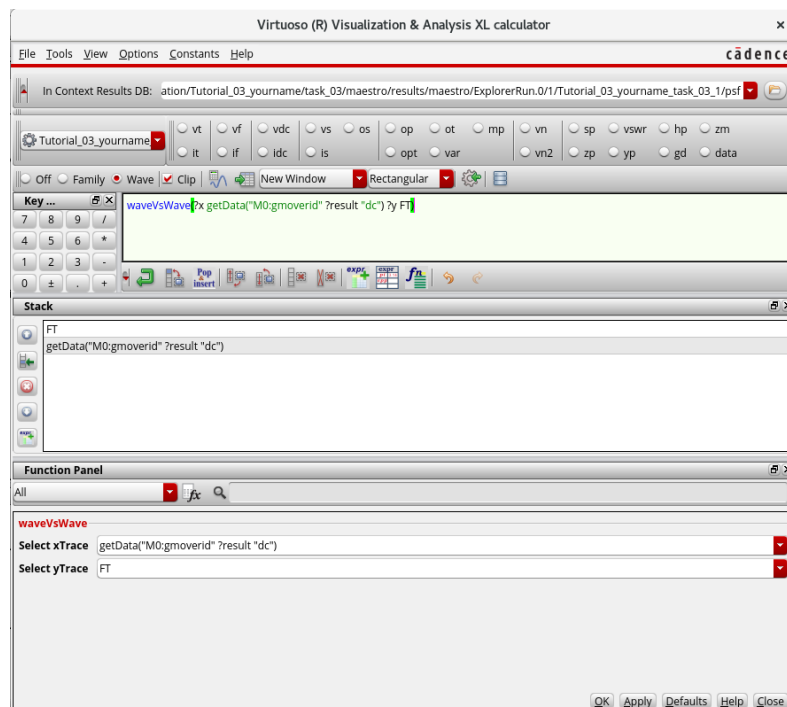
9. Notice that VGS is increased from 0 to 1, transistor transitions from weak to moderate inversion.
10. Next simulation will verify that as we move from weak to inversion to moderate inversion, the bandwidth will be maximize. Bandwidth will be represented by **transition frequency (Ft)**.
11. To simulate Ft, send the expressions of gm and cgg to calculator as they are required to form the expression of Ft as given by equation 1.



12. To calculate  $F_t$ , sent the expressions of  $G_m$  and  $C_{gg}$  to calculator and make the expression as given by equation 1. Once this expression is created, send it back to ADE.

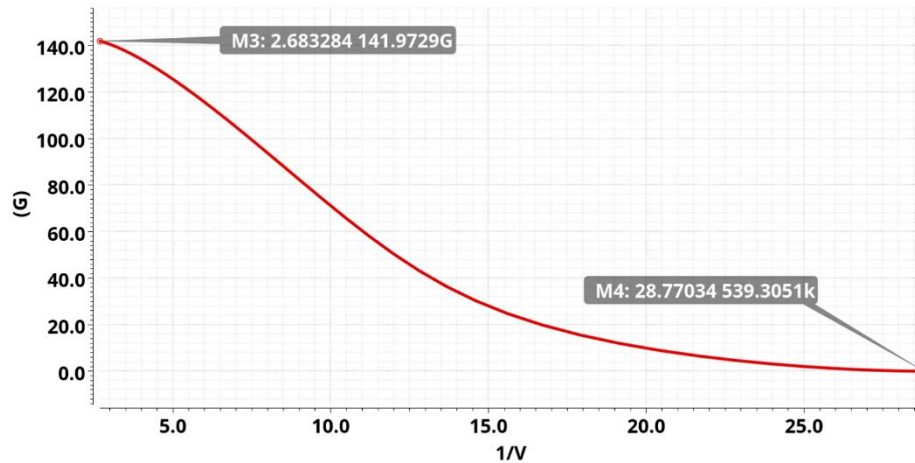


13. Next is to plot this expression of  $F_t$  against  $g_m/id$ . For that send both the expressions of  $F_t$  and  $g_m/id$  to calculator.
14. Use the **WaveVsWave** function from the calculator to plot  $g_m/id$  on x axis and  $F_t$  on y-axis as shown below. Send this expression back to ADE and plot it.





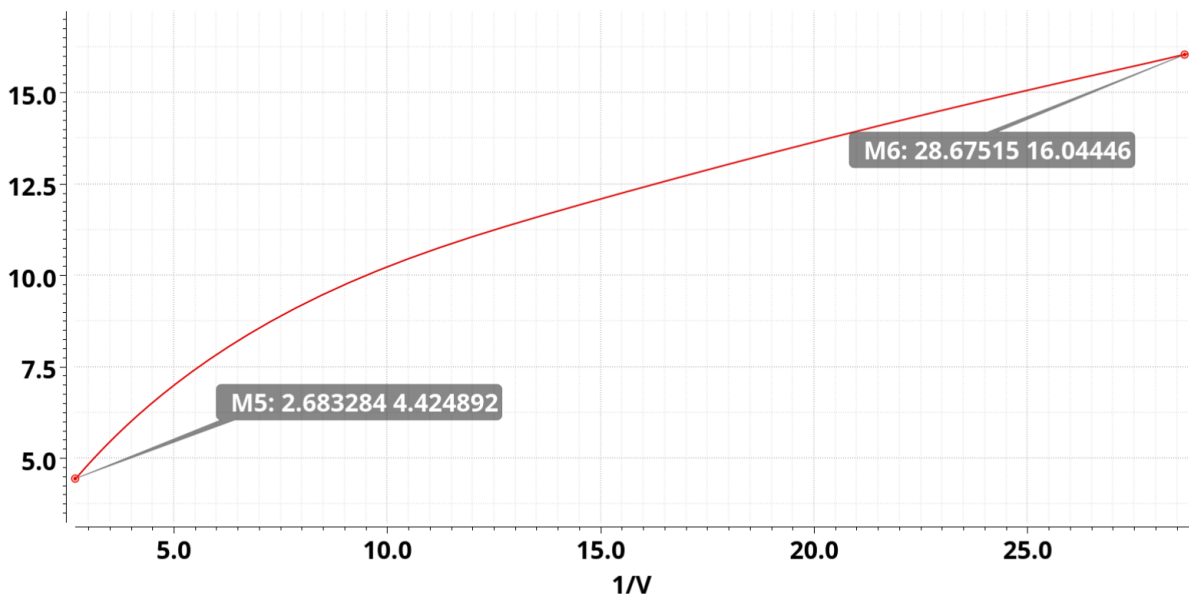
15. Plot of  $f_t$  against  $g_m/I_D$  will be plotted as shown below.



16. As it is clear from the graph that for lower value of  $g_m/I_D$  (strong inversion) maximum bandwidth can be achieved, however as we move to weak inversion (higher value of  $g_m/I_D$ ), bandwidth gets reduced.

17. Next simulation is to plot intrinsic gain or small signal voltage gain which is **transconductance times the internal resistance of the MOSFET**. It is our assumption that that since bandwidth is decreasing with increasing  $g_m/I_D$ , the gain must increase to keep the Gain and bandwidth product constant across the range.

18. To make the intrinsic gain expression fetch  $g_m$  and  $r_{out}$  from results browser and multiple them to make an expression for intrinsic gain using the calculator as done before for  $f_t$ . Then, plot this intrinsic gain against  $g_m/I_D$  as done for  $f_t$ . The curve will be plotted as shown below.



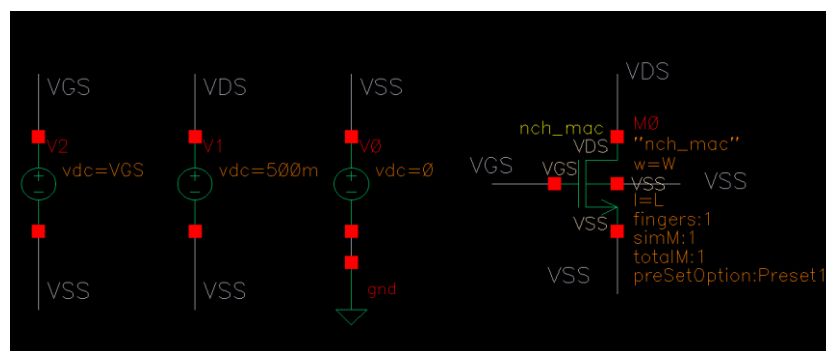
19. Evaluating intrinsic gain ( $A_v$ ) and transit frequency ( $f_T$ ) against  $g_m/I_D$  illustrates the fundamental **trade-offs** between **gain, speed, and power consumption in analog IC design**. Operating in weak inversion ( $g_m/I_D = 17.89\text{--}28.7\text{ V}^{-1}$ ) maximizes intrinsic gain and current efficiency at the expense of bandwidth. Conversely, biasing in strong inversion ( $g_m/I_D = 2.68\text{--}8.0\text{ V}^{-1}$ ) delivers maximum transit speed ( $f_T$ ), but degrades intrinsic gain and incurs high



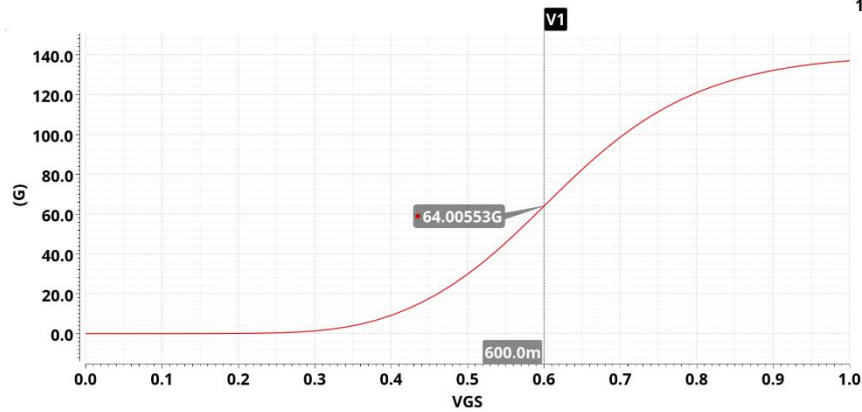
power dissipation. Moderate inversion ( $g_m/I_D = 8.0\text{--}17.89\text{ V}^{-1}$ ) provides the optimal design target, capturing near-peak intrinsic gain and strong high-frequency performance while maintaining high power efficiency. Consequently, weak inversion targets ultra-low-power biomedical sensors (< 10 kHz), strong inversion powers gigahertz-range RF front-ends, and moderate inversion serves as the default regime for general-purpose OTAs and data-converter drivers.

## B) Schematic Capture and Simulation for Transition Frequency ( $F_T$ ):

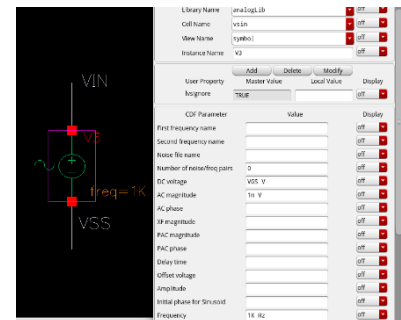
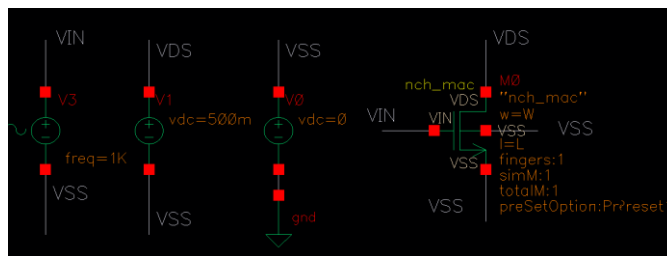
20. In the schematic editing tool window, in the function ribbon click on **(Create →Instance)** to add a circuit component or use the hot-key “i”
21. **Add Instance** window will appear click on **Browse** and **Component Browser** window will appear
22. In the Component browser from library section find and click on **tsmcN65** then from Cell column find and click on **nch** or **nch\_mac** and from the view section click on the symbol (note that on the Add Instance window various transistor parameters are shown. For now the default values will be used)
23. Now make the schematic as shown in the picture below:



24. After creating the schematic, save the schematic by pressing on the **check and save icon** and go to **Launch->ADE Explorer->Create New View**
25. Once the ADE window is open, first go to design variables and select **copy from cellview**, it will copy all design variables in the design to ADE explorer window. Then, go to analyses and define DC analyses and in the design variable select **VGS** and specify its sweep range from 0 to 1
26. Before simulating, go to **Setup->Simulation Files->Definition Files**. Use the same text file created for the previous lab called **save\_operating\_points.scs**
27. Now, run the simulation and the plot of  $I_D$  V/S  $V_{GS}$  will be plot. Next go to the results browser and under dc look for the following parameters:  **$g_m$  and  $c_{gg}$** .
28. Plot both, then send them to calculator and from the calculator make the expression as given in equation 01. Once the expression is created, send it to **ADE Outputs**. Now plot this expression and it will show a graph as shown below.



29. On the graph is plotted, place a marker at  $V_{GS}=600\text{mV}$ . A marker can be placed by using the shortcut key 'm'. By right clicking on the marker and clicking on the **point marker properties**, you can change the position of the marker to any value of x and y.
30. At  $V_{GS} = 600\text{mV}$ , the transistor achieves an FT of 64 GHz. Input signals exceeding this frequency will experience a current gain of less than 1 (0 dB), rendering the device incapable of current amplification.
31. Next is to verify this result using AC analyses. For that, modify the schematic by using a sinusoidal voltage source at the input, as given in the schematic below.



32. Then go to ADE explorer and from the **Analyses**, select AC analyses and use the following settings as given below.



Choosing Analyses -- ADE Explorer

Analysis

☐ tran ☐ dc ☒ ac ☐ noise  
☐ xf ☐ sens ☐ dcmatch ☐ acmatch  
☐ stb ☐ pz ☐ lf ☐ sp  
☐ envlp ☐ pss ☐ pac ☐ pstb  
☐ pnoise ☐ pxf ☐ psp ☐ qpss  
☐ qpac ☐ qpnoise ☐ qpxf ☐ qpss  
☐ hb ☐ hbac ☐ hbstb ☐ hbnoise  
☐ hbsp ☐ hbxf

AC Analysis

Sweep Variable

☒ Frequency  
☐ Design Variable  
☐ Temperature  
☐ Component Parameter  
☐ Model Parameter  
☐ None

Sweep Range

☒ Start-Stop Start  Stop   
☐ Center-Span

Sweep Type

Automatic

Add Specific Points ☐

Add Points By File ☐

Specialized Analyses

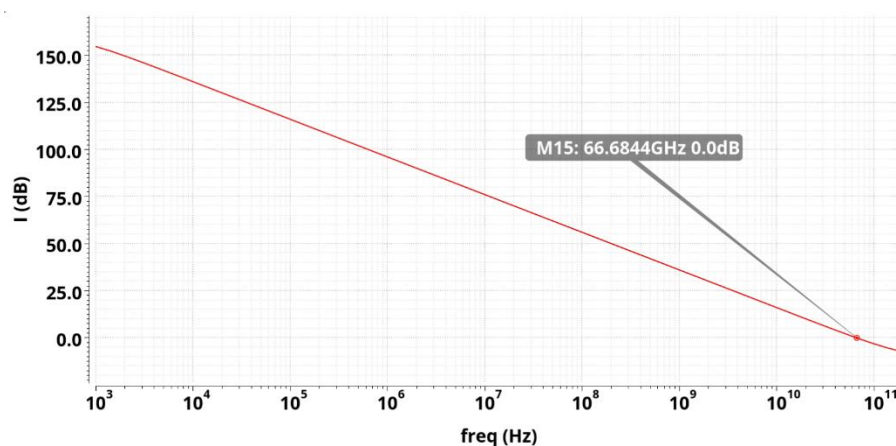
None

Enabled ☒

Options...

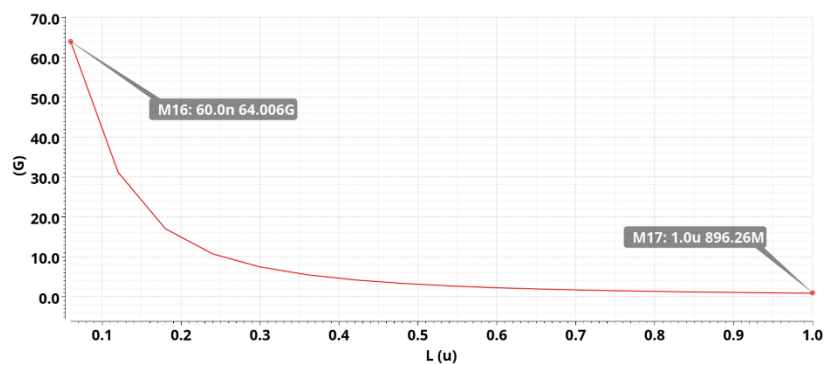
OK Cancel Defaults Apply Help

33. Set the value of VGS from design variable as 600mv and run only the AC simulation this time without selecting any outputs. Once the simulation is completed without any errors, go to **Results->Direct Plot->AC Gain & Phase**. Schematic window will open, click on the drain current terminal first and then click on the gate terminal. Note that the sequence of selecting the current terminal is important here.
34. A plot showing the magnitude in log scale and a phase will be shown. The phase curve is not required here so delete that only and keep the magnitude plot. The plot will look as shown below.





35. Place a marker at 0dB on the curve. The value of  $f_T$  from this method is 66.6GHz. Notice that the AC analyses 0 dB intercept (66.6 GHz) is slightly higher than the DC calculated  $f_T$  (64 GHz at  $V_{GS} = 600\text{mV}$ ). This is expected! The DC formula given by equation (1) is a first-order approximation used for quick estimations. In contrast, the AC sweep accounts for high-frequency  $C_{gd}$  feedthrough, pushing the physical unity current gain intercept higher
36. Next simulations will again use the DC analyses therefore, choose the DC analyses again from the analyses section and make change the schematic back to the one as shown in figure 1.
37. Open the object properties of the MOSFET in the schematic by using the shortcut key 'q'. Use a variable 'L' instead of the value 60nm in length.
38. Go to ADE Explorer and copy this new variable into the **Design Variables** and assign a value of 60n to it. Next, from analyses, go to DC and under design variable select the variable L and in the sweep range specify a range of 60n to 1u with step size of 60n. Note that in this simulation,  $V_{GS}$  will be kept at 600mV.
39. Now plot the expression previously sent to ADE explorer in step 9 and the graph as shown below will be plotted.



40. Transit frequency ( $f_T$ ) drops sharply as channel length ( $L$ ) increases—from **64 GHz** at  $L = 60\text{ nm}$  down to **896 MHz** at  $L = 1\text{ }\mu\text{m}$ . Longer channels increase carrier transit time and gate capacitance ( $C_{gg}$ ), causing speed to scale down inversely ( $f_T \propto 1/L$  to  $1/L^2$ ).

#### Circuit Application Trade-offs:

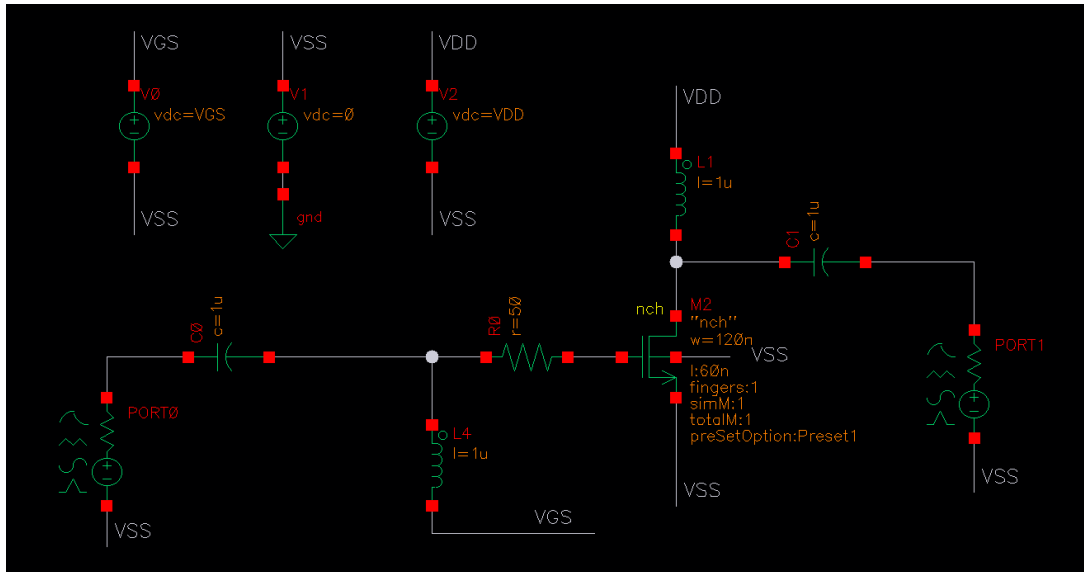
- **High-Speed / RF (60 nm):** Maximizes operating bandwidth (69.12 GHz) for high-frequency transceivers, LNAs, and fast switches.
- **Precision Analog (1  $\mu\text{m}$ ):** Trades speed (907.4 MHz) to achieve higher output resistance ( $r_o$ ) and intrinsic gain ( $g_m r_o$ ) in current mirrors and operational amplifiers.

41. Do the same process with width of the MOSFET. Keep length fixed at 60nm and plot transition frequency against width with a range of 0.2um to 1um. Show its plot and explain the results. What happens to transition frequency with increasing width while length remains fixed?
42. Show  $f_T$  plot against over-drive voltage. To do that plot  $V_{GS}$  and  $V_{TH}$  using the results browser. Send both voltages to calculator and make the expression for overdrive voltage as  $V_{GS} - V_{TH}$ . Now search for a function called '**WaveVsWave**' from calculator and use it to plot  **$f_T$  against  $V_{ov}$**  and explain the graph. How does transition frequency vary against  $V_{ov}$ ?



### C) Schematic Capture and Simulation for Fmax:

43. Next the simulation for Fmax is to be performed. Unlike  $f_T$ , which only measures short-circuit current gain,  $f_{max}$  accounts for layout parasitics like gate resistance ( $R_g$ ) to define the maximum frequency at which a transistor can deliver actual power gain
44. First create a new cell view in the same library. It is recommended to perform each task in a new cell view. Create the schematic as given below.



45. In the schematic, an instance called 'port' is placed at the gate and drain of the MOSFET. This component can be found in **analogLib**. The capacitor on the gate side is placed to pass only the AC signal from the port while blocking all DC components. The inductor is placed to prevent the AC signal from leaking into and corrupting the DC bias voltage that powers the MOSFET (remember from the previous lab that a proper gate bias voltage is required to turn on the transistor). Resistor **R0** models the gate resistance of the MOSFET. Since we are using a minimum-sized NMOS here, its intrinsic gate resistance is negligible, which yields an unrealistically high value for  $f_{max}$  (according to Equation 2).
46. At the drain side of the MOSFET, a similar combination is being used.
47. After successfully creating the schematic assign values of 1uF and 1uH to all the capacitors and inductors respectively.
48. Go to **launch-> ADE Explorer->create new view**. Once the ADE explorer is created, import all design variables to Maestro. Assign value of 1 to VDD and a value of 600m to VGS.
49. Next, go to analyses-> choose and select **sp**. Go to select and first select the input port and then select the output port. Under sweep variable select frequency and in range specify range of 1G to 2T with sweep type as



**Choosing Analyses -- ADE Explorer**

**Analysis**

- ☒ tran
- ☐ dc
- ☐ ac
- ☐ noise
- ☐ xf
- ☐ sens
- ☐ dcmatch
- ☐ acmatch
- ☐ stb
- ☐ pz
- ☐ lf
- ☐ sp
- ☐ envlp
- ☐ pss
- ☐ pac
- ☐ pstb
- ☐ pnoise
- ☐ pxf
- ☐ psp
- ☐ qpss
- ☐ qpac
- ☐ qpnoise
- ☐ qpxf
- ☐ qpdp
- ☐ hb
- ☐ hbac
- ☐ hbstb
- ☐ hbnoise
- ☐ hbsp
- ☐ hbxf

**DC Analysis**

Save DC Operating Point ☒

Hysteresis Sweep ☐

**Sweep Variable**

- ☐ Temperature
- ☐ Design Variable
- ☐ Component Parameter
- ☐ Model Parameter

Enabled ☒ **Options...**

**S-Parameter Analysis**

Ports **Select** **Clear**

/PORT0 /PORT1

**Sprobes**

| Port# | Name | Left Port | Right Port |
|-------|------|-----------|------------|
|       |      |           |            |

**Sweep Variable**

- ☒ Frequency
- ☐ Design Variable
- ☐ Temperature
- ☐ Component Parameter
- ☐ Model Parameter
- ☐ None

**Sweep Range**

- ☒ Start-Stop Start  Stop
- ☐ Center-Span

**Sweep Type**

☒ Logarithmic ☐ Points Per Decade

☐ Number of Steps

Add Specific Points ☐

Add Points By File ☐

**OK** **Cancel** **Defaults** **Apply** **Help**

- 
- The screenshot shows the Virtuoso® Studio application window. The 'Results' menu is open, displaying options: Plot Outputs, Direct Plot, Print, Annotate, Vector, Circuit Conditions..., Electrothermal Report..., EM/IR Data, Save..., Select..., Delete..., and Plotting/Printing Options... The 'Direct Plot' dialog box is also open, showing a list of plot types: Main Form..., Transient Signal, Transient Minus DC, Transient Sum, Transient Difference, AC Magnitude, AC dB10, AC dB20, AC Phase, AC Magnitude & Phase, AC Gain & Phase, Equivalent Output Noise, Equivalent Input Noise, Squared Output Noise, Squared Input Noise, Noise Figure, and DC. The 'Main Form...' option in the 'Direct Plot' dialog is highlighted with a red rectangle.



52. From the next window that appears, do the following settings

**Direct Plot Form**

Plotting Mode: Append

**Analysis**

☐ dc ☒ sp

**Function**

|                                       |                            |                                |                            |
|---------------------------------------|----------------------------|--------------------------------|----------------------------|
| <input type="radio"/> SP              | <input type="radio"/> ZP   | <input type="radio"/> YP       | <input type="radio"/> HP   |
| <input type="radio"/> GD              | <input type="radio"/> VSWR | <input type="radio"/> NFmin    | <input type="radio"/> Gmin |
| <input type="radio"/> Rn              | <input type="radio"/> rn   | <input type="radio"/> NF       | <input type="radio"/> Kf   |
| <input type="radio"/> B1f             | <input type="radio"/> GT   | <input type="radio"/> GA       | <input type="radio"/> GP   |
| <input checked="" type="radio"/> Gmax | <input type="radio"/> Gmsg | <input type="radio"/> Gumx     | <input type="radio"/> ZM   |
| <input type="radio"/> NC              | <input type="radio"/> GAC  | <input type="radio"/> GPC      | <input type="radio"/> LSB  |
| <input type="radio"/> SSB             | <input type="radio"/> Mu   | <input type="radio"/> Mu_prime |                            |

Description: Maximum Power Gain

**Modifier**

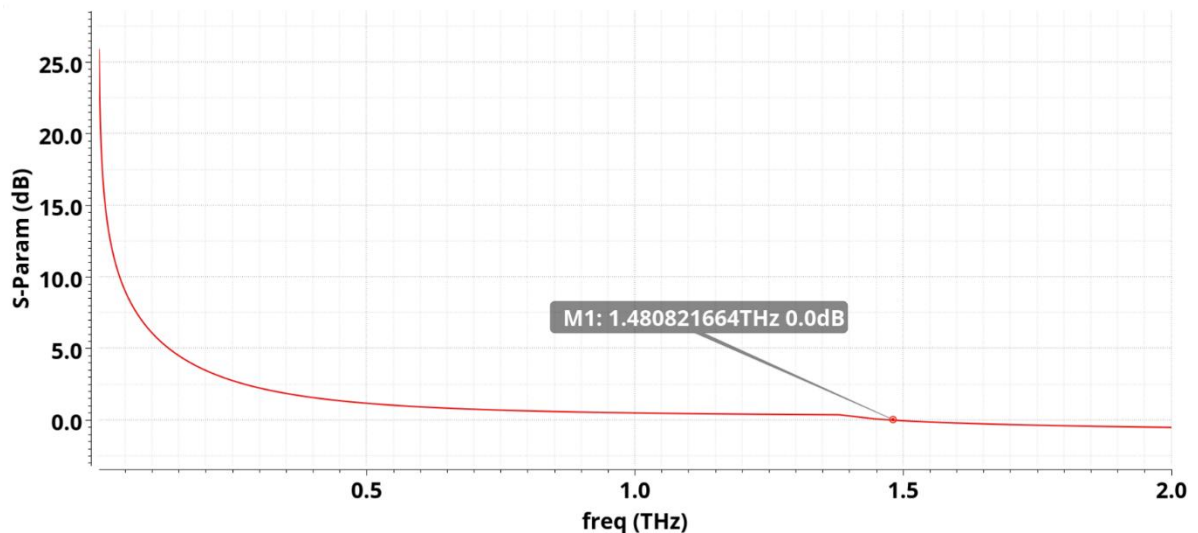
☐ Magnitude ☒ dB10

Add To Outputs ☒ **Plot**

> Press plot button on this form...

**Close** **Help**

53. Graph as shown in the picture below will be plotted. Place a marker at 0dB. The value at 0dB is the Fmax.

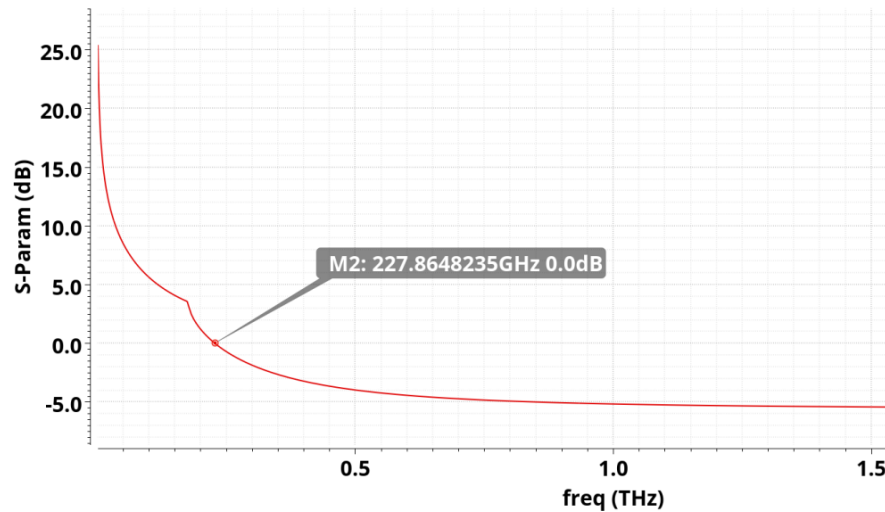


54. Note that the Fmax value is unrealistically high. The reason being the gate resistance RG is very low along with CGD at this device sizing.

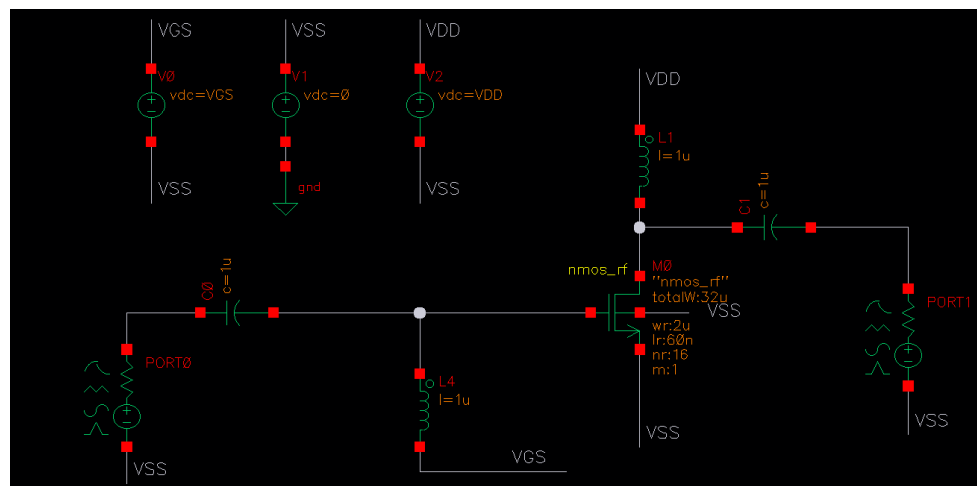
55. Next, change the size of the MOSFET to 5u/120n and perform the same simulation. Since in the last simulation, we checked the option to send the outputs to ADE, now we don't need to go to Results and select direct plot, once the simulation is done the graph will be displayed.



56. After increasing width to 5um, notice that now the Fmax value has dropped to **227GHz**.



57. Next simulate by changing the fingers of the MOSFET. Increase the number of finger and see the effect on Fmax. Does it decrease or increase? And why?
58. While we have extracted **Fmax** using the nch device up to this point, Fmax is most relevant to **RF circuit design** where dedicated nmos\_rf or pmos\_rf devices are used so now re-run the simulation by simply replacing the nch transistor with an nmos\_rf transistor as shown in the picture below



59. Note that we do not include an external RG resistor for this simulation, because the **nmos\_rf** model already accounts for intrinsic gate resistance.
60. Perform the same simulation as before and plot fmax. Change the number of fingers and observe the effect



## Summary

- $g_m/I_D$  & Inversion Regime Mapping: DC operating point sweeps of the NMOS device to map  $g_m/I_D$  and  $f_T$  against  $V_{GS}$ , establishing the boundaries for weak, moderate, and strong inversion.
- Speed, Gain, and Power Trade-off Analysis: Evaluation of intrinsic gain ( $A_{v0}$ ) and transit frequency ( $f_T$ ) against  $g_m/I_D$  to characterize the trade-off between power efficiency and the gain-bandwidth product across inversion regimes.
- Direct  $f_T$  Extraction: Dedicated high-frequency small-signal simulation to determine the exact unity-current-gain transit frequency ( $f_T$ ).
- $f_{max}$  S-Parameter Characterization: Two-port S-parameter analysis (sp) to extract the maximum oscillation frequency ( $f_{max}$ ) based on unilateral power gain.



## Exercise Activities

### Activity (3.1)

- Repeat the exercise done in para (A) for pmos transistor
  - I. Show all the graphs as done in para (A)

### Activity (3.2)

- For the PMOS transistor perform the following simulations
  - I. Vary the length of the transistor from 60n to 600n with step size of 60n and explore how does  $f_T$  changes. Show simulation results
  - II. Vary the width of the transistor from lowest size to 1u and explore the result on  $f_T$ .

### Activity (3.3)

#### Speed-Efficiency Figure of Merit (FOM)

$$FOM = f_T \cdot \left( \frac{g_m}{I_D} \right)$$

The Speed-Efficiency FOM quantifies the trade-off between bandwidth ( $f_T$ ) and power efficiency ( $g_m/I_D$ ), reaching a maximum peak in moderate inversion to reveal the optimal bias point.

- III. For NMOS transistor of size 120n/60m with  $V_{DS}$  of 500mV and  $V_{GS}$  0 to 1, plot the curve of FOM and identify the peak FOM.

### Activity (3.4)

- I. Plot the  $f_{max}$  for PMOS pch and for pmos\_rf. Compare the  $f_{max}$  of PMOS with that of the NMOS simulated in the para (c) .

#### Instructions for report:

The report must include the following:

- 1) All schematics clearly showing sizing and simulation results for guided tasks and for the exercise tasks
- 2) The structure the report as follows: Introduction->objectives->guided tasks  
-> exercise tasks.